Term 4 Checklist: Forces and Motion

Topic	Learning Outcomes	RAG	
		1	2
Forces and Motion	Describe the difference between contact and non contact forces		
	State a range of contact and non-contact forces		
	State which variable goes on each axis		
	Draw force arrows on objects that are at rest or moving		
	Calculate the resultant force of an object when two forces are acting on it		
	Explain why an object will start to accelerate using the idea of forces		
	Explain why an object will travel at a steady speed using the idea of forces		
	State the units of mass and weight		
	Explain why your weight changes on different planets but your mass does not		
	"Calculate your mass or weight. Try this one: Earth's gravity is 9.8N/kg, Mr Wormstone has a mass of 78kg. What is his weight?"		
	State the forces that occur when a boat moves through water and floats on water		
	State useful and unhelpful examples of friction		
	Explain why friction occurs between two surfaces		
	Describe the difference between the independent and dependent and control variables		
	State the equation that links speed, distance and time.		
	State the units of speed, distance and time		
	"Calculate speed, distance or time using the equation s = d/t. Try this one: A car travels 90m in 45s. What is its speed?"		
	Describe factors that affect stopping distances		
	Describe the difference between braking distance and thinking distance		
	Explain the difference between speed (or velocity) and acceleration		
	Recall and use a formula to calculate acceleration		
	Explain how pressure is caused inside a can of coke in terms of particles		
	Use the idea of forces why a rocket can take off		
	Explain how more aerodynamic shapes help rockets fly		